SA 220 AQUACULTURE

PRODUCTION FACILITIES

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CAGES

- There are different types of fish culture facilities and the most common ones are;

1. Cages
2. Pens
3. Tanks
4. Race ways
5. Recirculation aquaculture systems
6. Fish ponds

CAGES

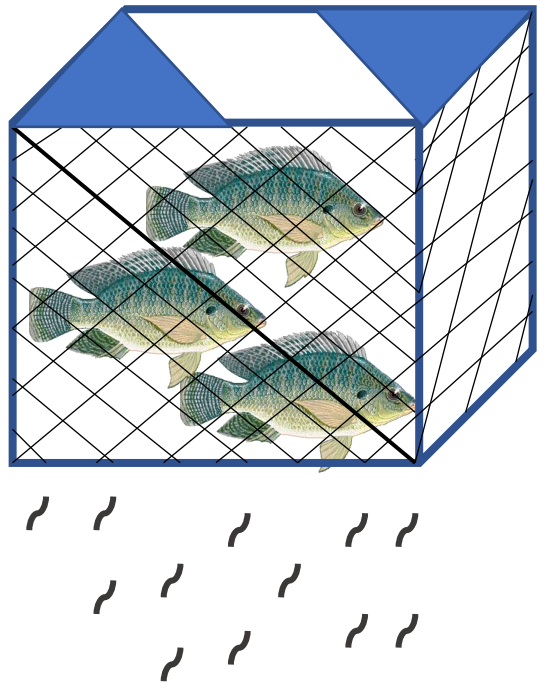
- Cages are an intensive production facilities.
- They are placed in natural water bodies after appropriate site selection.
- Stocking densities can be up to 53–160 kg/m^3

CAGES

- Floating cages may be small and of limited strength or they may be many thousands of cubic metres in volume.
- Designed for use in the open conservation dam, lake or ocean.







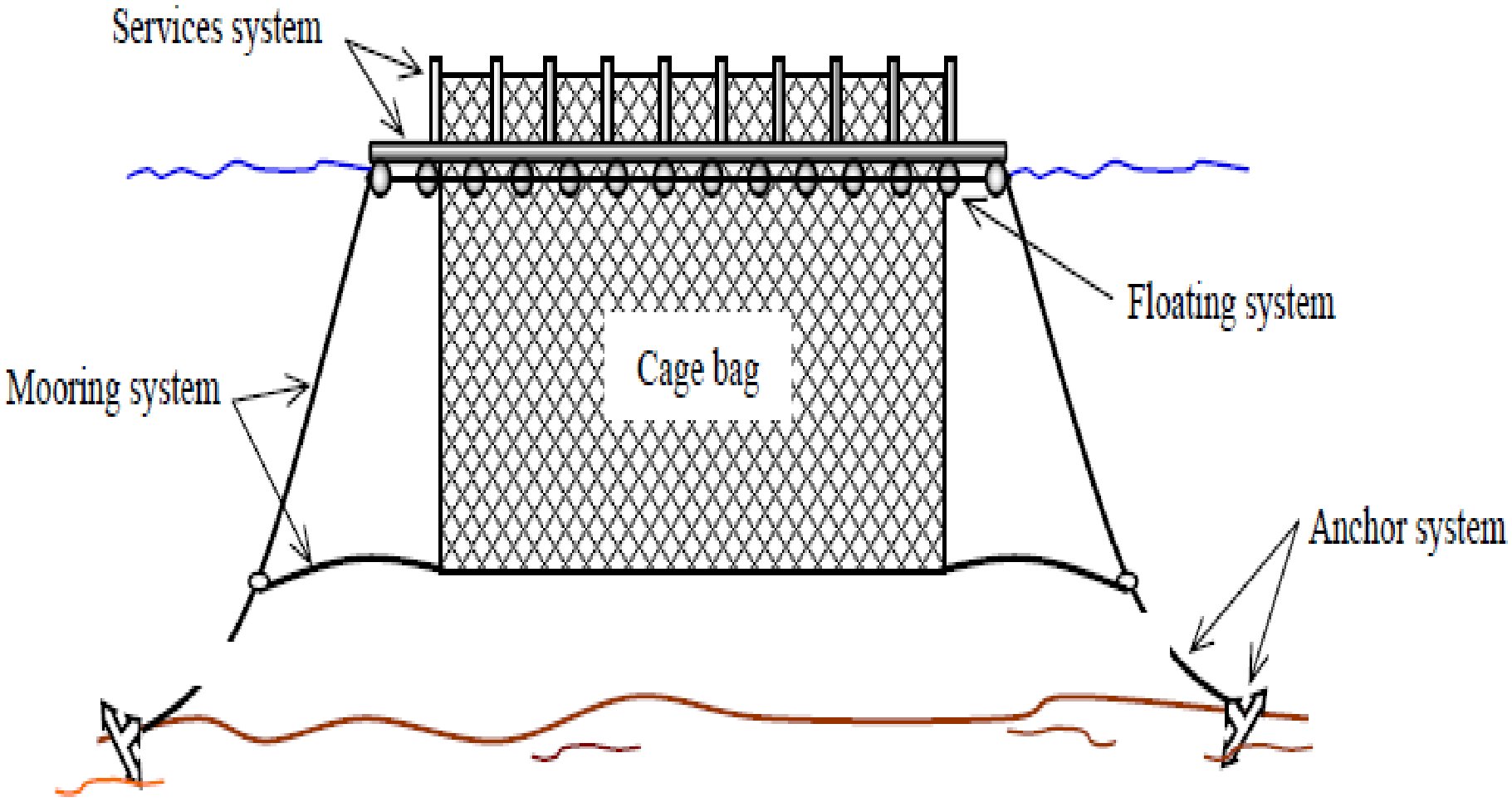
CAGES

- Cages located in open systems allow no control over water quality.

- Therefore require a good site with adequate exchange of high-quality water.



Principal components of a cage



Principal components of a cage

1. Floating system

- Provides buoyancy and holds the system at a suitable level in the surface of the water.
- In some cages this component is an important part to hold the shape of the cage.



Principal components of a cage

2. Services system

- This is the system required for providing operating and maintenance services, for example: feeding, cleaning, monitoring or grading



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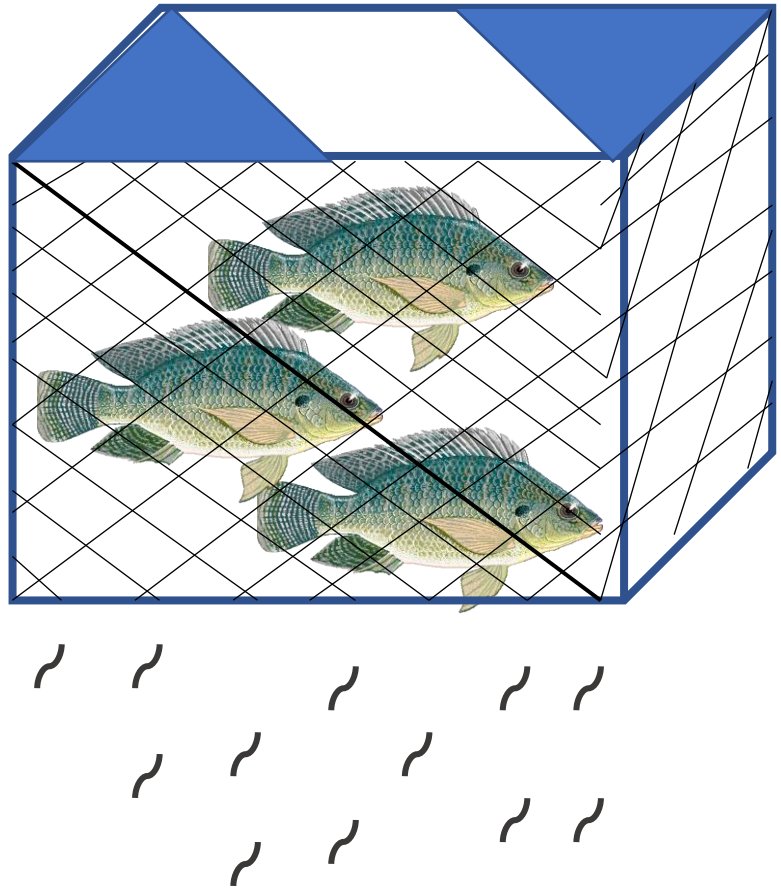
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Principal components of a cage

3. Cage bag

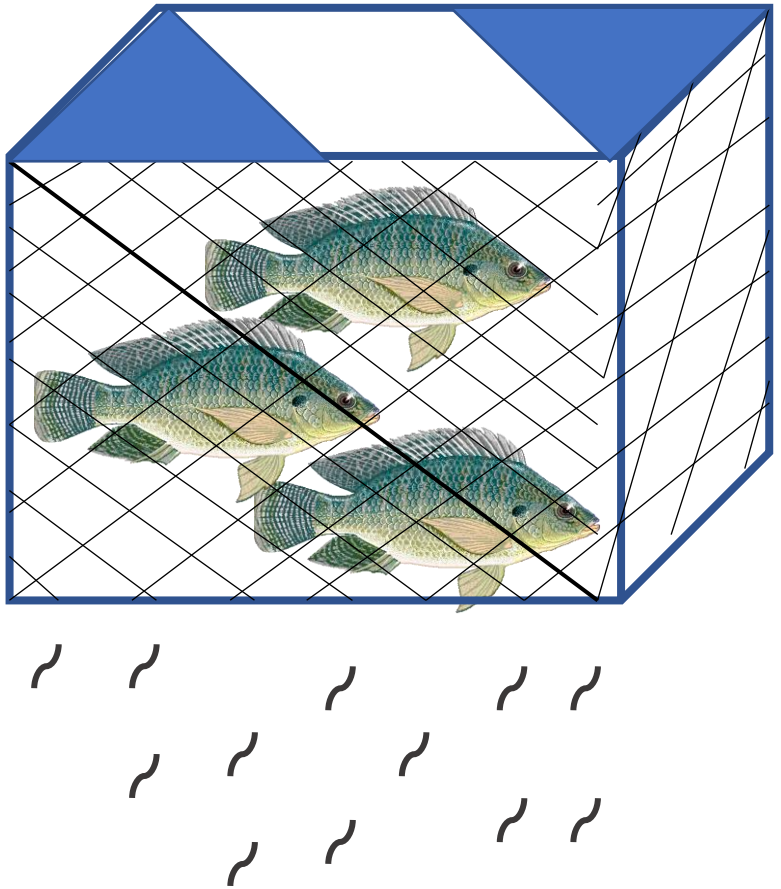
- The function of the bag is to contain and protect the fish.
- The nets are kept stretched vertically with weights at the bottom of the cage or fastened by rope (Chua and Tech 2002)



Principal components of a cage

4.Mooring system

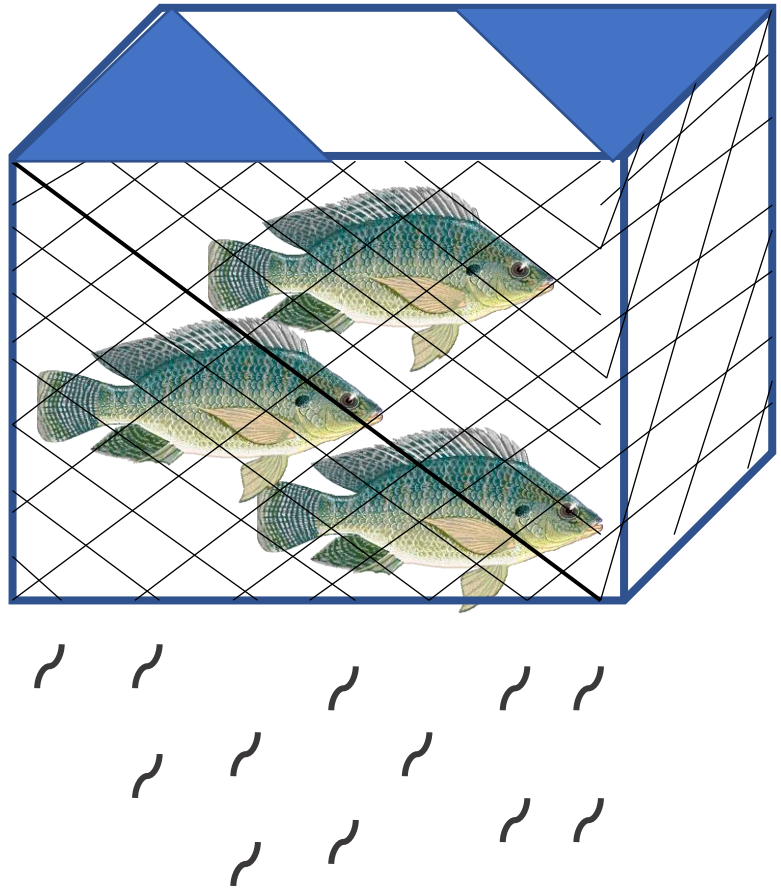
- This holds the cage in the suitable position according to the direction and depth decided in the design, and sometimes helps to maintain the shape of the cage.
- The materials used in the mooring systems are sea steel lines or chains.



Principal components of a cage

5. Anchor system

- This holds the cage and all the components in a particular site in the bottom and is connected to the cage by the mooring system.



ISSUES WITH CAGES

- Fouling of the meshes by species such as weeds which reduce the flow of water through the cage
- Predators, e.g. water monitors, seals and birds.
- Theft cases
- Diseases, which are difficult to manage, as cultured stock are difficult to observe and diseases are not easily treated except with medicated feeds
- Can potentially contribute to eutrophication (algal blooms).

Discussion

What are the advantages and disadvantages of cage fish farming?

FISH PENS

- Pens, enclosures or **hapas** are used in shallow water, typically in ponds, to create a restricted environment for culture of fish.

- They are not usually large, being in the order of tens of square metres or less.



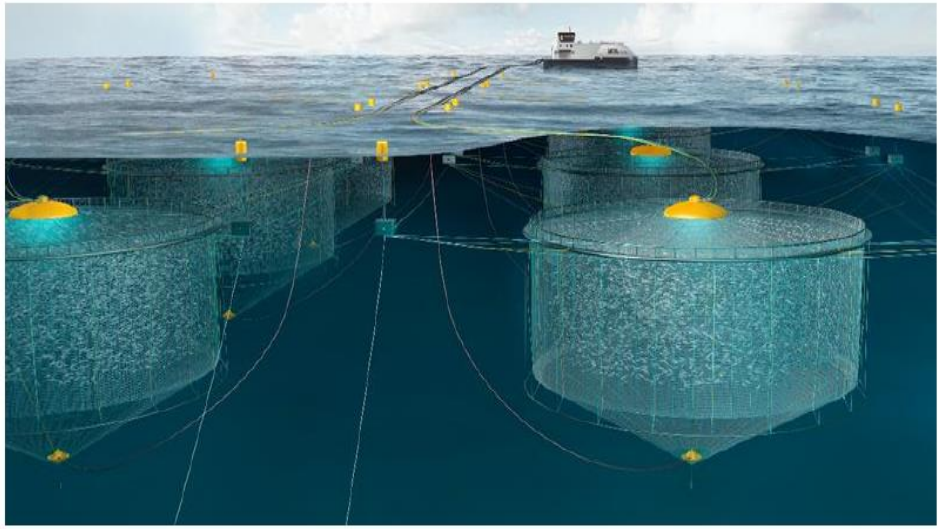
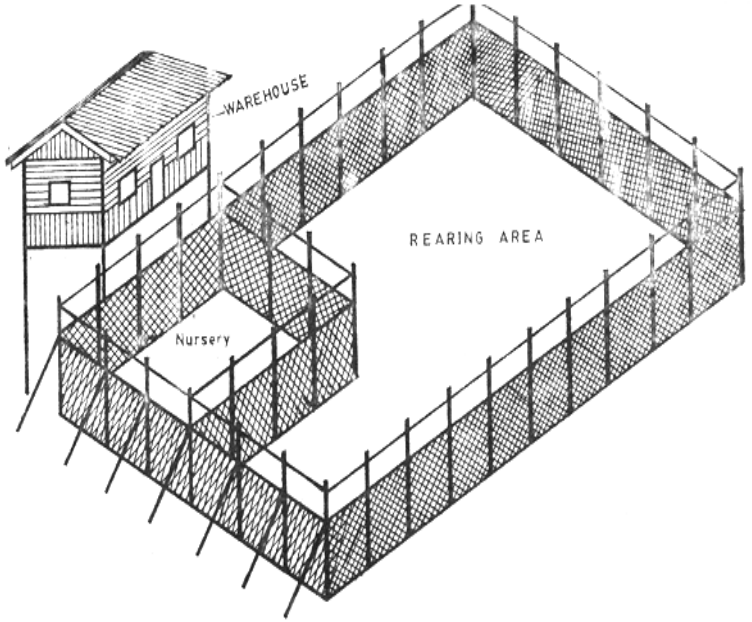


Figure 4: Sanford's Proposed Submersible Pen.

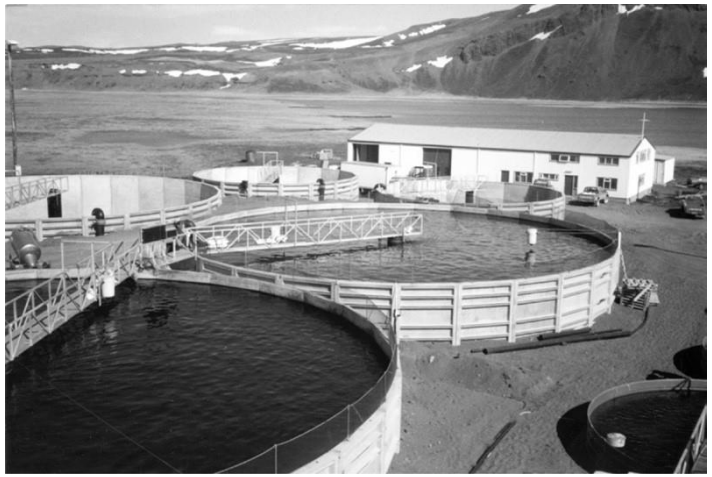
FISH PENS

- The walls of the enclosures may be closely spaced stakes, such as bamboo stems or wire and other mesh.
- This system of culture is practised mainly in developing countries.



TANKS

- Generally situated above ground on a solid base and may be used indoors or outdoors
- Have a wide range of dimensions & sizes, corresponding to their use e.g. various life cycle stages of fish & invertebrates.
- Range in size from tens of litres to hundreds of cubic metres



TANKS

- Rectangular, square, octagonal or circular tanks, lined with concrete or plastic, are also used in intensive tilapia farming.
- Stocking densities and yields are as high as those in raceway systems;
- Tank systems are commonly used in conjunction with a water recirculation system to minimise water use.



RACEWAYS

- Raceways are basically elongated tanks in which water enters at one end and exits at the other.
- They generally consist of elongate, narrow and shallow systems with continuous water flow.
- The limited cross-section of the raceway together with the strong flow rate are designed to keep continuous flow.
- It is a particularly suitable system for some fish, such as salmonids, that live in shallow streams and swim against the current.

RACEWAYS

- Raceways generally require large amounts of water per unit volume of the system.
- Freshwater raceways, the most common form, need to be sited near a freshwater spring or a substantial stream that does not dry up.



RACEWAYS

- As with other aquaculture, they need an unpolluted source of water.
- A potential problem with raceways is the deterioration of water quality along their length
- Raceways are also used for culturing high-value fish.
- The majority of raceways are flow-through

Recirculating Aquaculture System (RAS)

- Recirculating systems are usually characterised by minimal connection with the ambient environment and the original water source
- Have minimal exchange of H**2**O during a production cycle, hence the description as ‘closed’ systems
- The cost of construction and production in intensive RAS is high for commercial grow-out production

Recirculating Aquaculture System (RAS)

- However, the possibility of high yields with year-round production & proximity to market.
- Particulate matter (Settleable, suspended and fine waste solids) in the system resulting from feed and faecal matter
- Nitrogenous wastes (un-ionised ammonia- NH_3-N, ionised ammonia- NH_4-N, nitrite- NO_2-N and nitrate- NO_3-N

Recirculating Aquaculture System (RAS)

- Dissolved gases (O_2, CO_2 and N_2)
- Pathogens
- pH

PONDS

- Ponds are the most common culture facility used globally

- Several types of fish ponds

1. Earthen ponds
2. Pond-lined;
3. Semi-concrete ponds;
4. Concrete ponds.

EARTHEN PONDS

- Earthen ponds are impoundments for holding aquatic species
- It is the oldest Aquaculture structure because of the simplicity of basic pond culture in freshwater
- Generally a light algal bloom is maintained



EARTHEN PONDS

- Generally cheaper to construct per unit area than **tanks** and **cages**;
- May be inexpensive to run, depending on pumping costs.
- Cheap simple ponds are the most widely used freshwater & brackish water Aquaculture systems;



EARTHEN PONDS

- Have the **lowest stocking densities of the culture structures**; however, density varies according to whether the system is **extensive, semi-intensive and intensive.**



EARTHEN PONDS

- Reliable supply of good-quality water (preferably gravity fed)
- Relatively impermeable soils for construction e.g. clay-loam soil



EARTHEN PONDS

- Well-structured soils with good organic matter content to support pond ecosystems;
- Gravity drainage.



PONDS WITH LINERS

- Similar to earthen ponds as they are impoundments for holding aquatic species;
- Difference is that they have a plastic lining.



PONDS WITH LINERS

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- Difference is that they have a plastic lining.
- Soils that cannot hold water or lack of sufficient water (borehole)



SEMI-CONCRETE PONDS

- Similar to earthen ponds as they are impoundments for holding aquatic species
- Difference is that they have a concrete in the sides of the pond while the bottom is earthen.
- They are good for species that like burrowing like catfish

CONCRETE PONDS

- Can be constructed on any soil type
- Source of water can be borehole
- Constructed using blocks and cement
- No slopes



CONCRETE PONDS

- No fertilization is needed
- Concrete ponds are relatively expensive to construct compared to earthen ponds
- Requires regular flashing of water due to algae bloom.



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END

THANK YOU FOR YOUR ATTENTION.